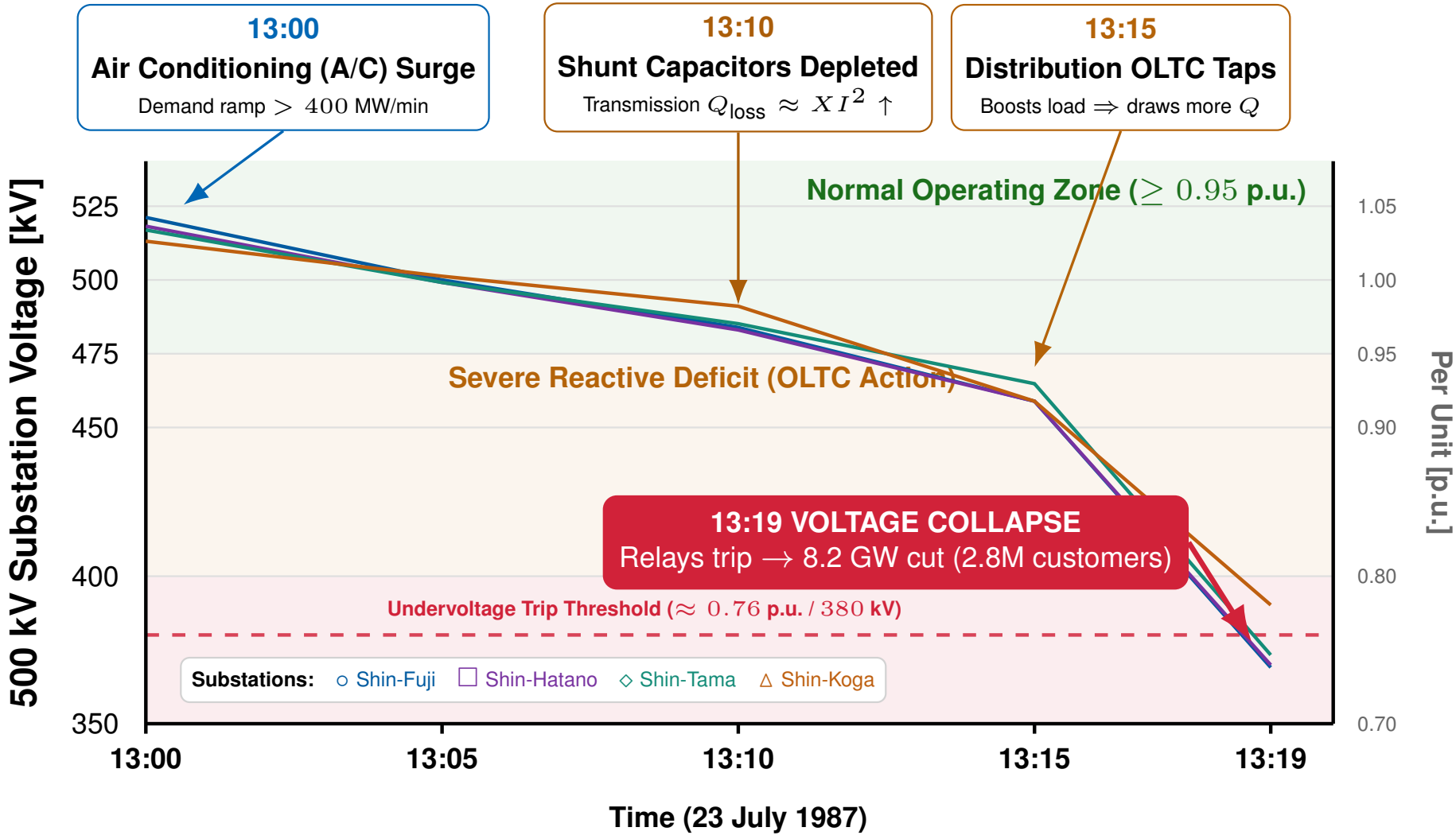

13:00
Air Conditioning (A/C) Surge
Demand ramp $>$ 400 MW/min
13:10
Shunt Capacitors Depleted
Transmission $Q_{\text{loss}} \approx XI^2 \uparrow$
13:15
Distribution OLTC Taps
Boosts load $\Rightarrow$ draws more Q
Normal Operating Zone ($\geq$ 0.95 p.u.)
Severe Reactive Deficit (OLTC Action)
13:19 VOLTAGE COLLAPSE
Relays trip $\rightarrow$ 8.2 GW cut (2.8M customers)
Undervoltage Trip Threshold ($\approx$ 0.76 p.u. / 380 kV)
Substations: ○ Shin-Fuji □ Shin-Hatano ◇ Shin-Tama △ Shin-Koga
500 kV Substation Voltage [kV]
525
500
475
450
400
350
Per Unit [p.u.]
1.05
1.00
0.95
0.90
0.80
0.70
13:00
13:05
13:10
13:15
13:19
Time (23 July 1987)